

Desktop Intelligent Micro-Ecosystem

TECHNICAL DOCUMENTATION



Version 1.0

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1. Product Overview

The Desktop Intelligent Micro-Ecosystem is an integrated desktop ecological system that combines indoor humidification, turtle aquaculture, and air purification functions. Through automatic control technology and intelligent design, this product provides users with a convenient, aesthetically pleasing, and sustainable micro-ecological environment while embodying green development principles and improving indoor environmental quality.

Project Objectives

- Create a living micro-world on the desktop featuring plants, turtles, and humidified air
- Minimize daily maintenance requirements through intelligent automation
- Enable voice interaction for enhanced user experience
- Promote sustainable development and environmental consciousness

Final Deliverable

A fully functional desktop ecosystem housed in a 225mm x 135mm x 195mm acrylic container, featuring automatic humidity control, voice-activated lighting and humidification, and a thriving environment for both plants and aquatic life.

2. Key Features

2.1 Multi-Function Integration

The system integrates multiple functions into a single compact unit:

- Indoor air humidification via ultrasonic mist generation
- Turtle aquaculture with dedicated aquatic and terrestrial zones
- Air purification through carnivorous and ornamental plants
- Ambient lighting for plant growth and aesthetic enhancement

2.2 Intelligent Control

- Automatic Control: Real-time temperature and humidity monitoring with automatic humidifier activation
- Voice Control: Chinese voice command recognition for hands-free operation
- Closed-Loop Feedback: Maintains optimal humidity levels (40%-60% RH) automatically

2.3 Customizable Design

Users can personalize their ecosystem by selecting from various plant species (mosses, carnivorous plants, ferns), arranging landscape elements (driftwood, stones), and customizing the 3D-printed turtle shelter dimensions.

2.4 Eco-Friendly Materials

- Primary container constructed from recyclable acrylic
- 3D-printed components use biodegradable PLA material
- All materials selected to minimize environmental impact

2.5 Easy Maintenance

- Modular, detachable design for convenient cleaning and component replacement
- Hinged lid design for easy water changes and plant care
- Clear maintenance protocols to ensure long-term system stability

3. Structure & Design

3.1 Container Structure

Primary Materials

Component	Material	Specification
Main Container	Acrylic Sheet	5mm thickness
Adhesive (Sealing)	Glass Glue	Waterproof sealant
Adhesive (Bonding)	Organic Solvent	For acrylic sheet joining

Structural Components

Main Tank: Overall dimensions of 225mm (L) x 135mm (W) x 195mm (H), houses the entire ecosystem and provides water reservoir for humidification.

Vegetation Zone: Vertical planting area for wall-mounted vegetation and horizontal planting area with layered substrate system for ground-level plants. Provides terrestrial habitat for plants and basking area for turtles.

Water Zone: Dimensions of 200mm (L) x 120mm (W) x 50mm (H), detachable module for easy water changes, serves as aquatic habitat for turtles and aquatic plants.

Lid Design: Two-part hinged lid connected with hinges, facilitates water changes and maintenance access, enhancing user experience through convenient operation.

4. Planting Scheme

4.1 Planting Substrate

Vertical Planting Substrate

Material	Thickness	Function
Vertical Planting Soil	30mm	Prevents plant detachment, enhances aesthetic value

Horizontal Planting Substrate (Layered System)

Layer	Material	Thickness	Function
Bottom	Vermiculite	20mm	Water permeability, nutrient provision
Middle	Dried Sphagnum Moss	10mm	Water permeability, prevents root rot
Top	Akadama Soil	30mm	Primary growing medium

4.2 Plant Species

Mosses

Small Gray Moss (xiao hui xian): Used for humidity retention and landscape layering.

Carnivorous Plants

Species	Quantity	Function
Venus Flytrap (bu ying cao)	1	Air purification, insect control
Pitcher Plant (zhu long cao)	1	Air purification, insect control
Sundew (mao gao cai)	1	Prevents flying insect proliferation

Ornamental Plants

Species	Quantity	Characteristics
Rabbit's Foot Fern (lang wei jue)	1	Shade-loving, moisture-loving
Maidenhair Fern (tie xian jue)	1	Shade-loving, moisture-loving
Podocarpus (luo han song)	3	Ornamental value

Nerve Plant - Forest Flame	1	Colorful foliage
Nerve Plant - Green Net	1	Variegated leaves
Air Plant - Yellow Flower	1	Epiphytic, low maintenance
Air Plant - Rubra	1	Epiphytic, decorative
Air Plant - Intermedia	1	Epiphytic, ornamental

Decorative Materials

- Driftwood (chen mu)
- Small stones (xiao shi zi)

4.3 Planting Procedures

Vertical Planting Procedure

0. Mix vertical planting soil with clean water until it reaches a thick, non-flowing consistency
1. Apply the mixture evenly to the inner wall of the container
2. Remove the backing paper from the moss sheet
3. Press the moss firmly onto the vertical soil layer
4. Mist with water to maintain moisture

Horizontal Planting Procedure

5. Spread vermiculite evenly across the bottom of the planting area
6. Layer dried sphagnum moss over the vermiculite (height adjustable as needed)
7. Add akadama soil on top until the substrate is completely covered
8. Plant angiosperms (flowering plants) in designated positions
9. Cover exposed soil with moss
10. Water thoroughly with settling water (ding gen shui)

4.4 Cultivation Conditions

Parameter	Requirement
Lighting	Diffused, indirect light; avoid direct sunlight
Temperature	Optimal: 15-28C; Minimum (Winter): Above 5C
Humidity	40%-90% RH; maintain via misting
Watering	Keep moist but avoid waterlogging

5. Water Area Design

5.1 Pool Specifications

Parameter	Specification
Material	Acrylic Sheet
Dimensions	200mm (L) x 120mm (W) x 50mm (H)
Design	Detachable module
Function	Turtle habitat and aquatic plant growth

Design Features

- Detachable Design: Allows for convenient water changes and cleaning
- Integration: Positioned within the main container for seamless ecosystem operation
- Land Access: Driftwood placement provides basking platforms for turtles

5.2 Turtle Shelter

Parameter	Specification
Manufacturing	3D Printing
Material	PLA (Polylactic Acid) - Eco-friendly biodegradable
Dimensions	100mm (L) x 100mm (W) x 50mm (H)
Customization	Dimensions adjustable based on turtle size

Functions

- Provides resting and hiding space for turtles
- Offers shade and security, reducing stress
- Enhances overall aesthetic appeal of the ecosystem
- Designed to allow free entry and exit for turtles

6. Control System

6.1 Hardware Components

Power Module

Parameter	Specification
Model	BD18650-1S-2P
Capacity	2400mAh
Output Voltage	Constant 5V
Dimensions	70mm x 38mm x 20mm
Connectors	Type-C Female + XH2.54 Male

Control Modules

Module	Specification
askpro Voice Recognition	Chinese voice commands, GPIO interface
2-Channel Relay	Control: 3.3V, Load: 5A
DHT11 Sensor	Temp: 0-50C, Humidity: 20%-95% RH

Actuator Modules

Parameter	Humidifier	Grow Light
Type	Ultrasonic	LED Four-Color
Voltage	DC 5V	5V
Current	300mA	-
Power	1.5-2W	-
Frequency	108kHz +/- 2kHz	-
Spectrum	-	400-700nm

6.2 Software Functions

Automatic Control

Temperature & Humidity Monitoring: DHT11 sensor continuously monitors environmental conditions. Data is transmitted to the control module via serial interface, enabling real-time feedback for responsive control actions.

Humidity Regulation Logic: Lower threshold at 40% RH, upper threshold at 60% RH. When humidity falls below 40% RH, the humidifier automatically activates until humidity reaches the target range. Hysteresis control prevents rapid cycling near threshold values.

Voice Control

Command (Chinese)	Function
jia shi (Add Humidity)	Activate humidifier
guan bi jia shi qi (Turn Off Humidifier)	Deactivate humidifier
kai deng (Turn On Light)	Activate grow light
guan deng (Turn Off Light)	Deactivate grow light

Command Execution Flow: (1) askpro module recognizes voice command, (2) Control signal sent to relay module, (3) Relay switches corresponding device on/off, (4) System confirms execution.

System Stability & Compatibility

- Hardware Compatibility: All modules connected via standard interfaces with high compatibility
- Software Stability: Program optimized for long-term continuous operation
- Error Handling: Software debouncing algorithm filters sensor noise

7. Installation & Usage

7.1 Installation Procedures

Step 1: Container Assembly

11. Assemble acrylic panels according to design drawings
12. Use glass glue for waterproof sealing at joints
13. Apply organic solvent for acrylic bonding
14. Install hinged lid, ensuring secure hinge attachment
15. Verify structural integrity and water-tightness

Step 2: Vegetation Planting

16. Prepare vertical planting area with vertical soil substrate (30mm)
17. Prepare horizontal planting area with layered substrate system
18. Plant moss, carnivorous plants, and ornamental plants according to design
19. Arrange decorative elements (driftwood, stones)
20. Water thoroughly to settle substrate

Step 3: Water Area Installation

21. Position detachable water pool in designated location
22. Place turtle shelter on the terrestrial area
23. Ensure turtle can freely enter and exit the shelter
24. Fill pool with appropriate water level

Step 4: Circuit Connection

25. Connect power adapter to power module (BD18650)
26. Connect voice recognition module, temperature/humidity sensor, and relay module per circuit diagram
27. Connect humidifier and grow light to relay module outputs
28. Verify all connections are secure and properly polarized

Step 5: System Testing

29. Power on system and verify all modules initialize correctly
30. Test voice command functionality
31. Test automatic control functionality

32. Verify DHT11 sensor reports accurate readings
33. Confirm system maintains target humidity range

7.2 Operating Instructions

Humidification Function

- Automatic Mode: System monitors humidity and activates humidifier when levels fall below 40% RH
- Manual Mode: User can speak "jia shi" to activate or "guan bi jia shi qi" to deactivate

Turtle Care

- Place turtle in water pool, ensuring free access to the shelter
- Change pool water weekly to maintain water quality
- Monitor turtle behavior and health regularly

Plant Maintenance

- Regularly inspect plant growth and health
- Prune dead or yellowing leaves as needed
- Adjust grow light duration based on plant requirements
- Maintain substrate moisture within recommended range

8. Maintenance & Precautions

8.1 Routine Maintenance

Water Pool Cleaning

- Frequency: Weekly water changes recommended
- Procedure: Remove detachable pool, drain existing water, clean with soft cloth (avoid abrasive materials), refill with fresh dechlorinated water

Plant Care

- Regularly inspect plants for signs of stress or disease
- Remove dead leaves promptly to prevent decay
- Adjust substrate moisture based on season and plant growth
- Trim overgrown vegetation to maintain desired landscape

Equipment Inspection

- Weekly: Verify electrical connections, check for loose wires, inspect humidifier output, verify grow light operation
- Monthly: Clean sensor surfaces, check power module charge capacity, inspect relay module switching

8.2 Precautions

34. Avoid Direct Sunlight: Prolonged sun exposure may damage plants and accelerate material degradation
35. No Corrosive Cleaners: Use only mild, non-abrasive cleaning agents
36. Regular Water Level Monitoring: Maintain appropriate water levels for turtle health
37. Environmental Conditions: Avoid placing in excessively humid or high-temperature environments
38. Electrical Safety: Ensure all connections are dry before powering system
39. Water Quality: Use dechlorinated water to protect both turtle and plants

9. Environmental Benefits

9.1 Air Purification & Humidification

Air Purification

- Carnivorous plants absorb harmful substances including formaldehyde and benzene
- Plant photosynthesis releases oxygen, improving indoor air quality
- Plant root systems help filter airborne particles

Humidification Function

- Ultrasonic Technology: Atomizes water into micron-sized particles (4-5 micrometers)
- Even Distribution: Mist spreads uniformly throughout the environment
- Health Benefits: Maintains comfortable humidity levels for respiratory health
- Ideal For: Dry climates and air-conditioned spaces

9.2 Biodiversity

Plant Diversity

Rich variety of species (mosses, carnivorous plants, ferns, angiosperms) increases ecosystem complexity. Different growth requirements and ecological functions create complementary relationships, enhancing ecosystem stability through biodiversity.

Animal Habitat

- Turtle shelter provides secure resting environment
- Turtle activities and metabolic byproducts provide nutrients for aquatic plants
- Creates closed-loop ecological cycle
- Adds interactive and educational element to the ecosystem

9.3 Sustainable Development

Eco-Friendly Materials

- Acrylic panels are recyclable
- PLA material used for 3D printing is biodegradable
- Plant and animal selections follow ecological balance principles
- No invasive species introduced

Sustainability Features

- **Energy Efficiency:** Automatic control reduces unnecessary power consumption
- **Reduced Maintenance:** Intelligent automation minimizes manual intervention
- **Extended Lifespan:** Detailed maintenance guidelines ensure long-term operation
- **Resource Conservation:** Modular design allows component replacement

10. Appendices

Appendix A: Circuit Diagram Description

The control system follows a modular architecture with the following connections:

Power Module (BD18650 5V) connects to: (1) askpro Voice Recognition Module via GPIO Interface, (2) DHT11 Temperature & Humidity Sensor via Digital Pin, (3) 2-Channel Relay Module (Control: 3.3V, Load: 5A), which in turn connects to Humidifier (Channel 1) and Grow Light (Channel 2).

Connection Notes: All modules share common ground. Signal lines use appropriate pull-up/pull-down resistors where required. Power distribution ensures adequate current capacity for all components.

Appendix B: Programming Code Overview

Automatic Control Logic

The system continuously reads DHT11 sensor data for temperature and humidity. When humidity falls below 40%, the relay activates the humidifier. When humidity exceeds 60%, the humidifier is deactivated. A software debounce delay filters sensor noise before making control decisions.

Voice Control Logic

The system listens for voice input through the askpro module. Upon recognizing commands such as "kai deng" (turn on light), "guan deng" (turn off light), "jia shi" (add humidity), or "guan bi jia shi qi" (turn off humidifier), the corresponding relay channel is activated or deactivated.

Software Debounce Implementation

To prevent erratic readings caused by mist interference, the system implements a software debounce algorithm. After reading the raw sensor value, a 500ms settling delay is applied before taking a confirmation reading. If the difference between readings is within an acceptable threshold, the value is returned; otherwise, the process repeats recursively.

Appendix C: Planting Record

Vertical Planting Record

Parameter	Data
Planting Method	Vertical planting
Substrate	Vertical planting soil

Plant Species	Small Gray Moss (xiao hui xian)
Substrate Thickness	30mm
Cultivation Condition	Keep moist

Horizontal Planting Record

Parameter	Data
Planting Method	Horizontal planting
Layer 1 (Bottom)	Vermiculite, 20mm
Layer 2 (Middle)	Dried Sphagnum Moss, 10mm
Layer 3 (Top)	Akadama Soil, 30mm
Light	Diffused light, avoid direct sun
Temperature	15-28C (minimum 5C in winter)
Humidity	40%-90% RH

Appendix D: Product Demonstration Video

For a complete demonstration of product functionality including voice control examples, automatic humidity regulation, and system operation, please refer to the product demonstration video.

Video Contents: Voice command demonstration ("kai deng", "guan deng", "jia shi", "guan bi jia shi qi"), automatic humidity control in action, turtle behavior in the aquatic environment, plant growth and ecosystem overview, system assembly highlights.

Desktop Intelligent Micro-Ecosystem

Technical Documentation

For full technical details, you can read our project documentation [\[link to PDF\]](#)

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